

=> Regression Analysis:

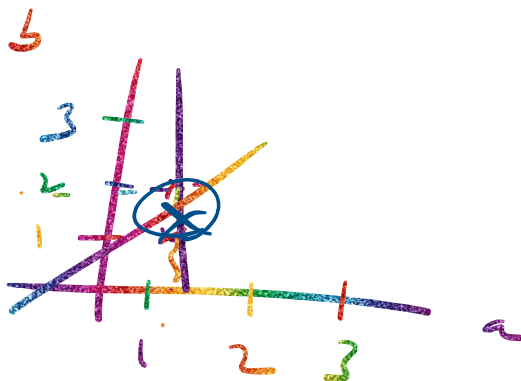
=> Standard Error: $S_{xy} = \sqrt{\frac{SSE}{n-2}}$

=> SSE : Sum of Squared Errors

=> Testing the hypothesis about the slope & Correlation Co-efficient.

=> t-Test for the slope:

When $y = a + bx$; if $b = 0$ then model cannot be considered as linear.



$$\begin{aligned} y &= a + bx \\ a &= 1 \\ b &= \cancel{x} \ 0 \\ x &= 2 \end{aligned}$$

$\therefore H_0 : \beta = 0$ against $H_a : \beta \neq 0$

$$t_c = \frac{\hat{\beta}}{\sqrt{\frac{SSE}{(n-2) \cdot SSX}}}$$

$$SSX = \sum (x_i - \bar{x})^2 \quad [\hat{\beta} = 0.744]$$

y	$\hat{y}$	Error $y_i - \hat{y}_i$	Squared Error $(y_i - \hat{y}_i)^2$
60	59.37	0.63	0.39
68	69.78	-1.78	3.16
75	77.2	1.8	3.24
80	82.43	-2.43	5.90
85	83.18	1.82	3.31

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$SSE = 0.39 + 3.16 + 3.24 + 5.90 + 3.31$$

$$SSE = \underline{16}$$

$$SSX = \underline{\underline{716.8}}$$

$$T_c = \frac{b}{\sqrt{\frac{SSE}{(n-2) \cdot SSX}}} = \frac{0.744}{\sqrt{\frac{1.4}{3 \times 716.8}}}$$

$$T_c = \underline{\underline{8.625}}$$

cum. prob	t _{.50}	t _{.75}	t _{.80}	t _{.85}	t _{.90}	t _{.95}	t _{.975}	t _{.99}	t _{.995}	t _{.999}	t _{.9995}
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300

[illegible]

$$y = a + b_1 x_1 + b_2 x_2$$

$$\Rightarrow a = \bar{y} - \{ (b_1 \cdot \bar{x}_1) + (b_2 \cdot \bar{x}_2) \}$$

$$\begin{array}{c} \swarrow \quad \searrow \\ \frac{SS_{xy}}{SS_{xx}} \quad \frac{SS_{xy}}{SS_{xx}} \end{array}$$

$\Rightarrow$ Least Squares Estimators: Multiple Regression:

$$y = a + b_1 x_1 + b_2 x_2 + \dots + b_k x_k$$

$$b = \begin{pmatrix} a \\ b_1 \\ b_2 \\ \vdots \\ b_k \end{pmatrix}$$

$$LSE = (X^T \cdot X)^{-1} \cdot X^T \cdot y$$

x_1 Product 1	x_2 Product 2	y Weekly Sales
1	1	1
2	5	6
3	8	8
4	2	12

$$X = \begin{pmatrix} \text{Bias} & x_1 & x_2 \\ 1 & 1 & 1 \\ 1 & 2 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 2 \end{pmatrix}$$

$$LSE = (X^T \cdot X)^{-1} \cdot X^T \cdot y$$

$$LSE = (X^T X)^{-1} X^T y$$

$$X^T = \begin{pmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 5 & 8 & 2 \end{pmatrix}$$

(4x3) (3x4)

$$X^T X = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 5 & 8 & 2 \end{pmatrix} \times \begin{pmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 10 & 19 \\ 10 & 30 & 46 \\ 19 & 46 & 109 \end{pmatrix}$$

(3x4) (4x3) 3x3

$$(X^T X)^{-1} = \begin{pmatrix} 5.15 & -0.55 & -0.30 \\ -0.55 & 0.20 & 0.016 \\ -0.30 & 0.016 & 0.054 \end{pmatrix}$$

$$(X^T X)^{-1} X^T = \begin{pmatrix} 5.15 & -0.55 & -0.30 \\ -0.55 & 0.20 & 0.016 \\ -0.30 & 0.016 & 0.054 \end{pmatrix} \times \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 5 & 8 & 2 \end{pmatrix}$$

3x3 3x4

$$= \begin{pmatrix} 0.05 & 0.47 & -1.02 & 0.19 \\ -0.32 & -0.298 & 0.155 & 0.26 \\ -0.065 & 0.005 & 0.185 & -0.125 \end{pmatrix}$$

$$\{(X^T X)^{-1} X^T\} \cdot y = \begin{pmatrix} 0.05 & 0.47 & -1.02 & 0.19 \\ -0.32 & -0.298 & 0.155 & 0.26 \\ -0.065 & 0.005 & 0.185 & -0.125 \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ 6 \end{pmatrix}$$

$$(X^T X)^{-1} \cdot X^T y = \begin{pmatrix} 0.05 & 0.47 & -1.02 & 0.19 \\ -0.72 & -0.278 & 0.155 & 0.26 \\ -0.065 & 0.005 & 0.185 & -0.125 \end{pmatrix} \times \begin{pmatrix} 1 \\ 6 \\ 8 \\ 12 \end{pmatrix}$$

$\underbrace{\quad}_{3 \times 4} \quad \underbrace{\quad}_{4 \times 1}$

$$= \begin{pmatrix} -1.65 \\ 3.48 \\ -0.05 \end{pmatrix} \begin{matrix} a \\ b_1 \\ b_2 \end{matrix}$$

$$\hat{y} = a + b_1 \cdot x_1 + b_2 \cdot x_2$$

$$= -1.65 + 3.48 \cdot x_1 + (-0.05) \cdot x_2$$

$$x_1 = 5 \quad \text{and} \quad x_2 = 3$$

$$y = -1.65 + 3.48 \times 5 + (-0.05 \times 3)$$

$$\underline{\underline{y = 15.56}}$$

$$\begin{matrix} [1] & + & [5, 3] \\ \downarrow & & \downarrow \quad \downarrow \\ a & & b_1 x_1 \quad b_2 x_2 \end{matrix}$$

$$[1 \times 3] \times \begin{bmatrix} \quad \\ \quad \\ \quad \end{bmatrix} \begin{matrix} 3 \times 1 \\ \quad \end{matrix}$$

$$y = _ + _ + _$$

⇒ R^2 - Score - Goodness of Fit

$$R^2 = \frac{SSR}{SST} \quad \text{or} \quad 1 - \left(\frac{SSE}{SST} \right)$$

⇒ Adjusted R^2 :

$$R_{adj}^2 = 1 - \frac{SSE / (n - k - 1)}{SST / (n - 1)}$$

F-Test (ANOVA) for Regression :

ANOVA	D-F	SS	MSS	F-Ratio
Regression	1	SSR	$MSR = SSR$	$F_c = \frac{MSR}{MSE}$
Residual	$n - 2$	SSE	$MSE = \frac{SSE}{(n - 2)}$	
Total	$n - 1$	SST		

⇒ t-test for correlation co-eff :

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

H_0 : Correlation coeff is blv Δy is zero

H_1 : Not zero.

$$t_r = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}} \quad (n-2) - \text{Dof}$$

$\Rightarrow$ Confidence Interval Estimate of Slope:

$$\hat{b} \pm t_{\alpha/2} \times \sqrt{\frac{SSE}{(n-2) \cdot SSX}}$$